

Hyrax

A low-code solution for rapid experimentation with ML in astronomy

Drew Oldag - UMN ML Workshop Nov 2026

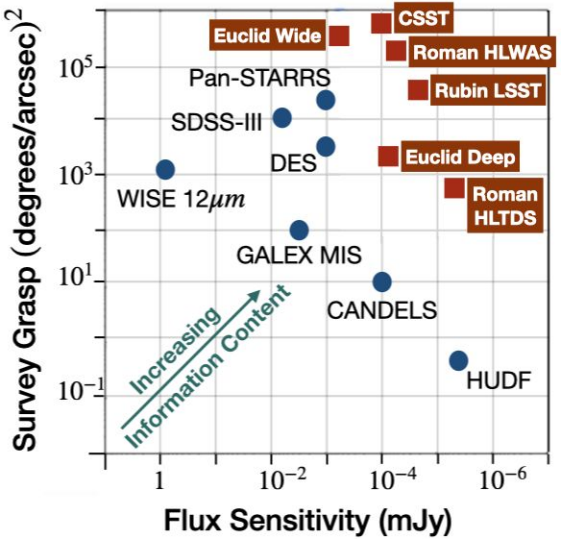
Carnegie Mellon University

LSST Discovery Alliance

Schmidt Sciences

Motivation - scale and bottlenecks

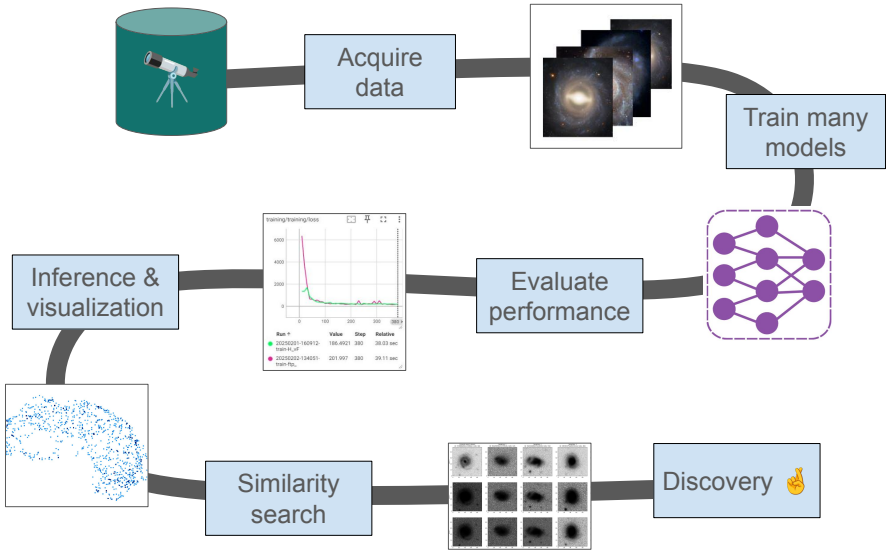
Survey scale



Ghosh, Oldag, & Tauraso et al. (in prep.)

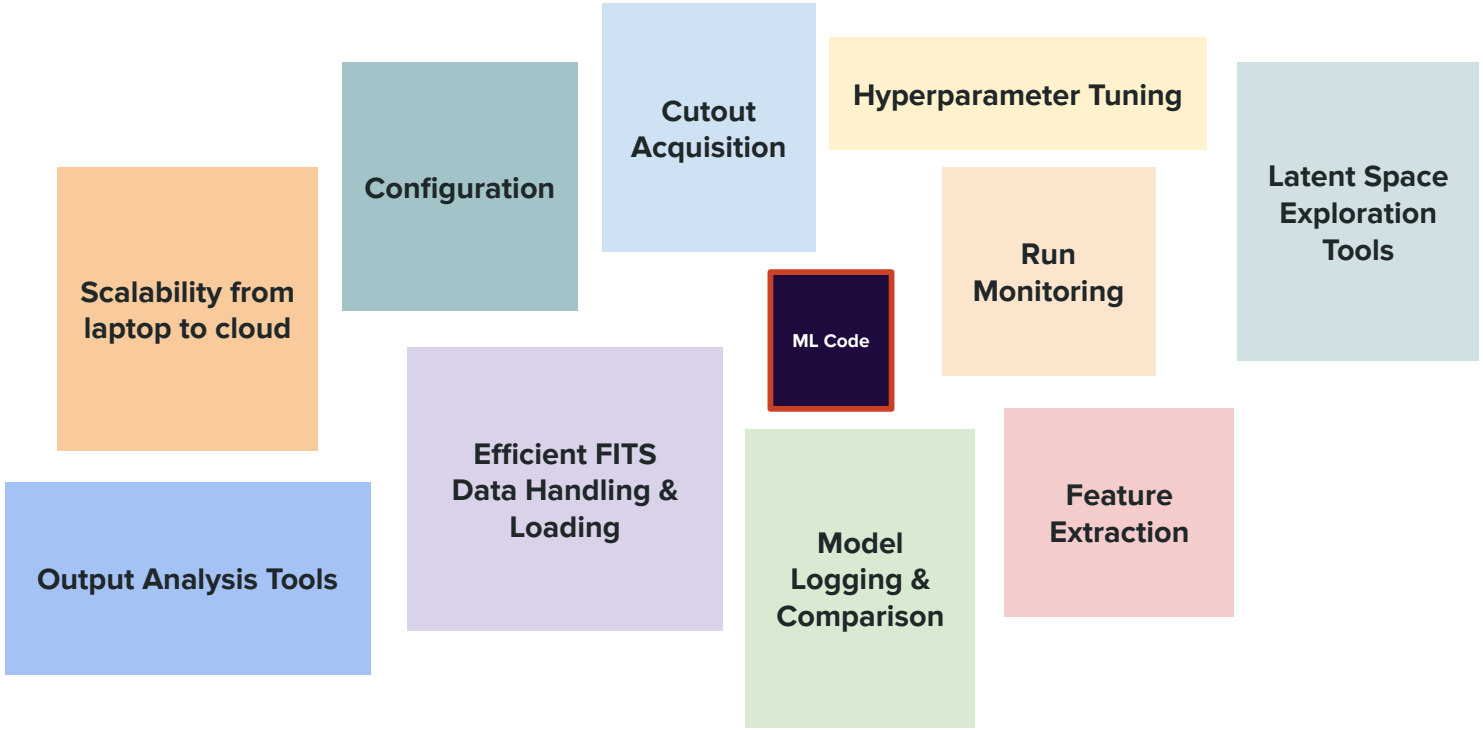
Numerous large surveys will generate a scale of data that conventional techniques will struggle to process.

Bottlenecks



Machine learning workflows in astronomy struggle with many bottlenecks - slow data acquisition, model training and comparison, visualization, etc.

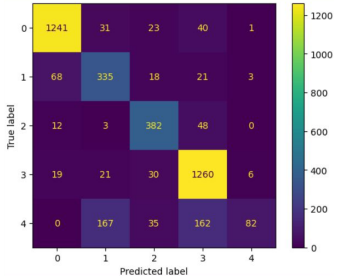
Machine learning is more than just ML



Inspired by D. Sculley et al. (2015)

Science results using Hyrax

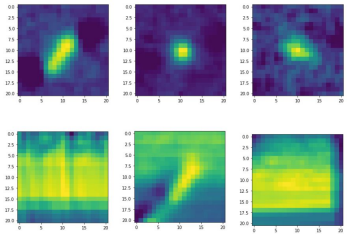
Lightcurve alert classification



- Follow up recs
- Supervised
- Multimodal: time series, spectra, images

A . Sasli, F. Fontinele Nunes

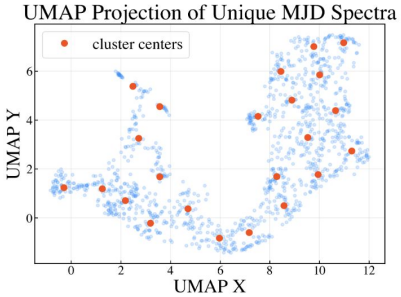
Small body classification



- Solar system science
- Supervised
- Image-based

M. West, A. Connolly

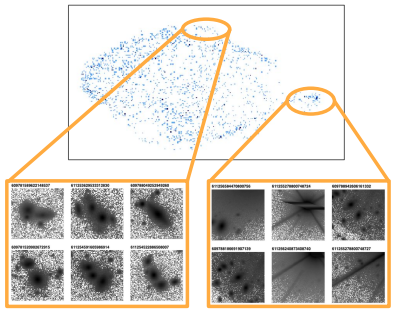
Spectra-based SN classification



- SN science
- Supervised, clustering
- Spectra-based

M. Dai, L. Cunningham

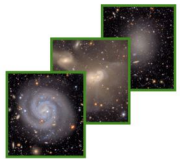
Image-based unsupervised discovery

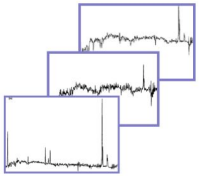


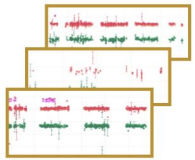
- Extragalactic science
- Unsupervised
- Image-based

Ghosh, Oldag, & Tauraso et al. (in review.)

Hyrax API







Model

Data

Configuration

HYRAX



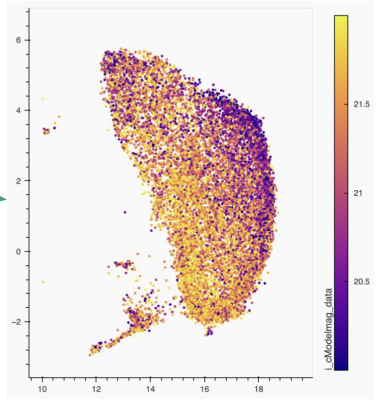
h.train()

h.infer()

h.visualize()

20260327-120406-train-Mc5D
- checkpoint_18_trainer_loss=-1.3244.pt
- checkpoint_epoch_20.pt
- runtime_config.toml
- tensorflow_metrics.local
- trained_weights.pth

20260327-120510-infer-ZRen
- results.lance
- inference_weights.pth
- runtime_config.toml
- tensorflow_metrics.local



Hyrax data model



Example: Alerts saved as .json

Each file might contain:

- Cut outs for image, psf, mask
- Many columns of metadata
- Previous observations including
  - Image cutouts
  - A few metadata columns

```
{
    "image": [[...]],
    "psf": [[...]],
    "mask": [[...]],
    "metadata": {
        "obj_id": <>,
        "ra": <>,
        "dec": <>,
        ...
    }
    "prev_obs": [
        {"image": [[...]], "metadata": {...}},
        ...
    ]
}
```

Hyrax data model



The dataset class receives the index and field from Hyrax and returns the appropriate data.

LSST-specific dataset can be used to efficiently produce image cutouts.

```python
class AlertDataClass(HyraxDataset):
    def __init__(self, data_location):
        self.alerts = sorted(
            os.list_dir(data_location)
        )

    def get_image(self, index):
        alert_file = self.alerts(index)
        alert = json.loads(alert_file)
        # ideally return numpy
        return alert['image']


    def get_psf(self, index):
        # ...
    def get_mask(self, index):
        # ...
```

Hyrax data model



The user tells Hyrax what data to work with via a “data request” configuration.

Hyrax uses this to load the correct dataset class and load only the required data fields.

```
data_request = {
    "train": {
        "data": {
            "dataset_class": "AlertDataClass",
            "data_location": "./data",
            "fields": ["image"],
        }
    },
    "validate": {
        "data": {
            "dataset_class": "AlertDataClass",
            "data_location": "./data",
            "fields": ["image"]
        }
    },
}
```

Hyrax data model



The `prepare_inputs` function is the connector between the data and the model.

It's purpose is ensure the shape and content of the data matches what the model needs.

It allows us to decouple the dataset class from the model.

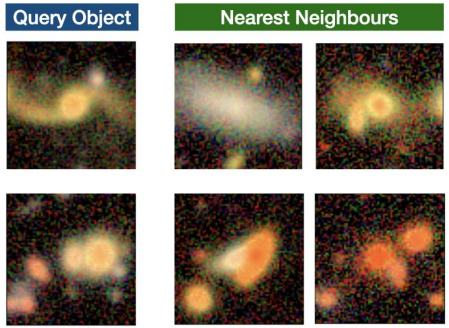
```python
@staticmethod
def prepare_inputs(data_dict) -> tuple:
    import numpy as np

    data = data_dict["data"]
    image = np.asarray(data["image"], dtype=np.float32)

    return (image, label)
```

Vector database and Model evaluation

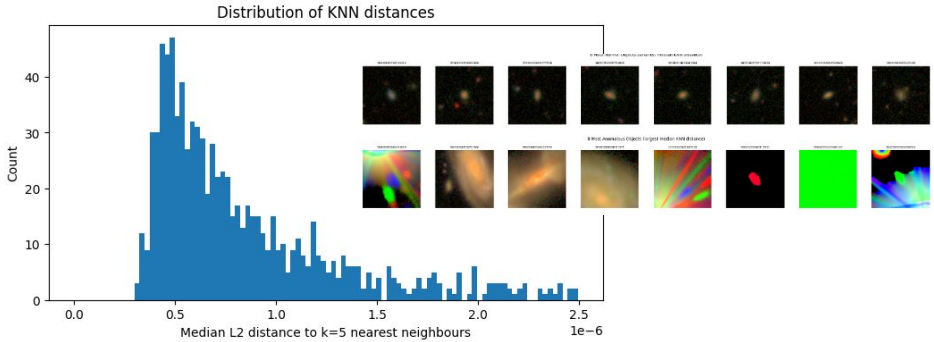
Efficient reverse search



Ghosh, Oldag, & Tauraso et al. (in review.)

Automatic metric tracking

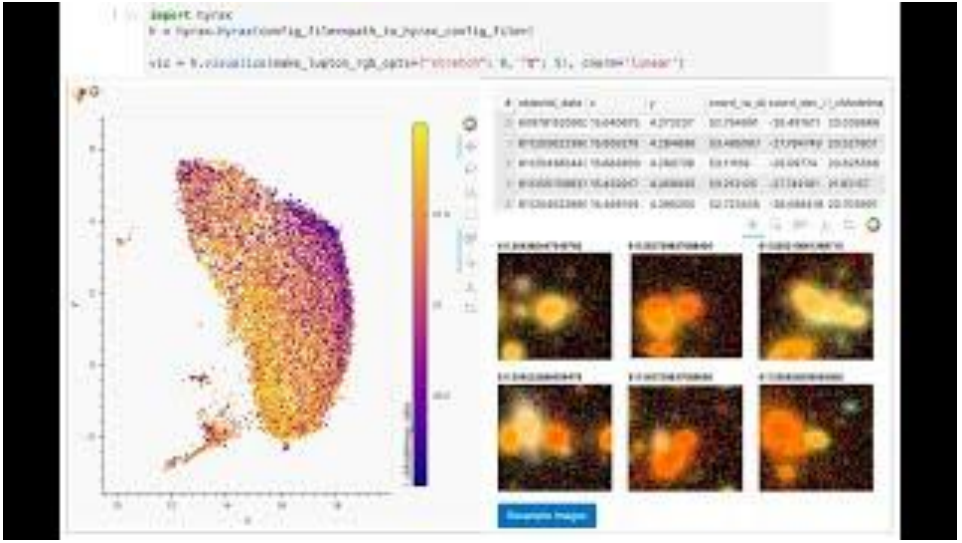




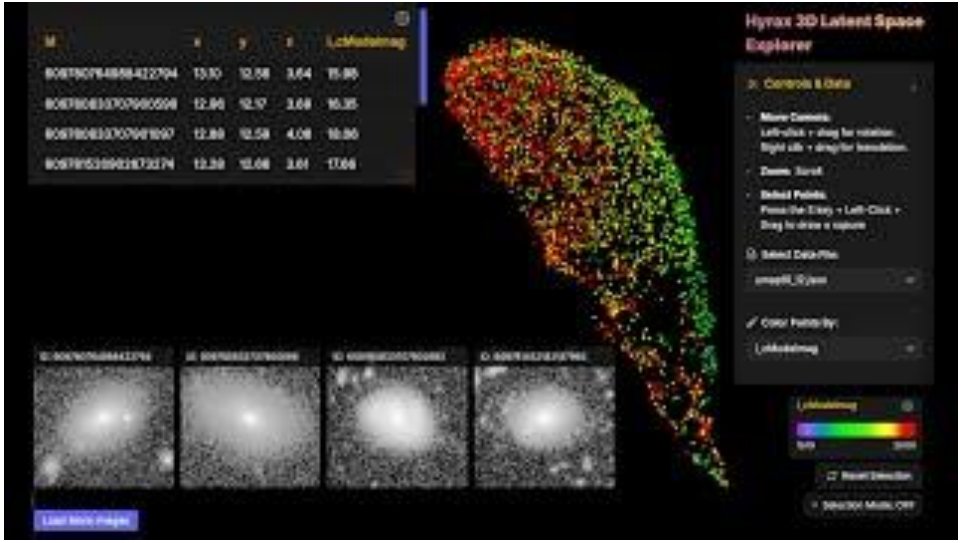
- Model perf metrics
- User defined model metrics
- Resource usage (GPU usage)
- Hyrax metrics (cache hit rate)

Interactive visualization

Jupyter notebook, 2D data

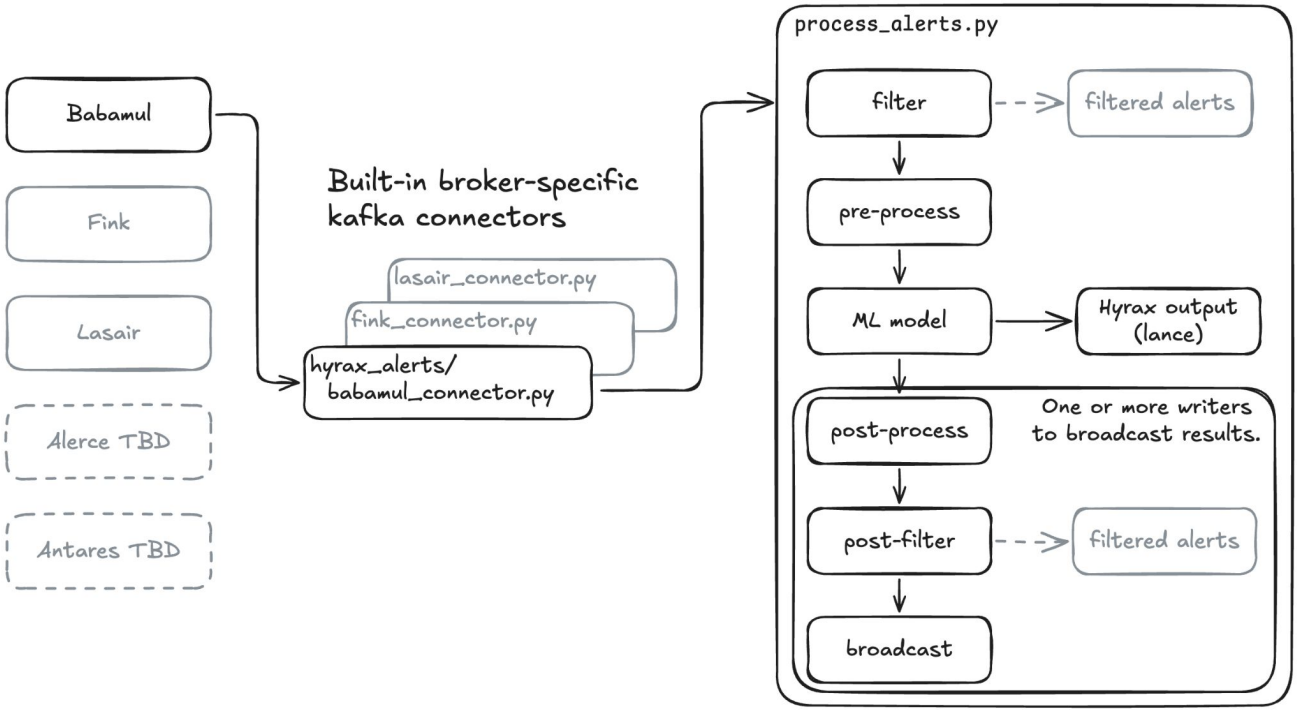


Browser based, 3D data



- Supports common interactions for data exploration
- Displays O(millions) of data points

Streaming datasets with Hyrax Alerts



https://github.com/lincc-frameworks/hyrax_alerts

Beta - HATS support LSDB-ML (LSDB+Hyrax)

Training - LSDB *then* Hyrax

```
import lsdb
import hyrax

tess_cat = lsdb.open_catalog("path/to/catalog")

h = Hyrax()

LSDBStreamDataset.register_catalog("my_cat", tess_cat)
my_data = {
    "dataset_class": "LSDBStreamDataset",
    "data_location": "lsdb://my_cat",
    "primary_id_field": "ticid",
}

h.set_config("data_request.train_stream.data", my_data)
h.set_config("model.name", "HyraxTs2Vec")

with h.train_stream() as session:
    for batch, metrics in session:
        # monitor training
```

Example training notebook

Inference - LSDB-ML wraps Hyrax

```
import lsdb
import lsdb_ml
import pandas as pd

tess_cat = lsdb.open_catalog("path/to/catalog")

pred_meta = pd.DataFrame({"pred":pd.Series(dtype="float32")})

my_model_config = lsdb_ml.ModelConfig(
    model="HyraxModel",
    model_weights_file="./model_weights.pth",
    output_fields=pred_meta,
)

with lsdb_ml.ml_client(my_model_config):
    tess_pred = lsdb_ml.predict(tess_filtered_nans)
    tess_pred.compute()
```

Example inference notebook

More Ways to Engage



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Hyrax paper on ArXiv
https://arxiv.org/abs/2605.18959v1

Discovery Alliance channel
#lincc-frameworks-hyrax

Tech & Science Bi-Weekly Meeting
Zoom (time TBA in slack)

Documentation
hyrax.readthedocs.io

On GitHub
github.com/lincc-frameworks/hyrax